PART-A (Protein Engineering) | 15 marks | 1 mark per question

Fill in the blanks in each of the following questions.

1 1) Different amino acid sequences of similar length can form very similar three-dimensional (polypeptide backbone) structures if these sequences are quite similar in terms of the placement of mostly similar amino acids at comparable chain positions. (True/False) ~~[illegible]~~ True

1 2) In the example taught to you, the replacement of an enzyme's active surface by that of another homologous enzyme requires that the surface in question be constituted of beta strands and loops, but no alpha helical components. (True/False) True

1 3) In experimental terms, the rational approach to protein engineering differs from the combinatorial approach mainly in the number of mutants/variants that have to be processed on the route to producing a protein variant with desired qualities. (True/False) True

1 4) The rational approach to protein engineering can sometimes by supported by the computational approach and *vice-versa*. (True/False) True

1 5) As an approach, chemical mutagenesis (done to obtain desirable mutants of a protein) fall under which of the three approaches of protein engineering named under parentheses (combinatorial/rational/computational) combinatorial. Choose the most appropriate option.

6) Protein hydrophobic core reengineering aids in the alteration of protein aggregation (thermal stability/aggregation/degradation). Choose the most appropriate option.

1 7) Protein electrostatics reengineering aids in the alteration of protein thermal stability (thermal stability/aggregation/degradation). Choose the most appropriate option.

1 8) Protein domain/substructure fusion is usually done in order to add structural/functional features to an existing protein. (True/False) True

9) Disulphide bond introduction to prevent domain dissociation can prevent protein aggregation through intermolecular domain associations. (True/False) False

1 10) Five of the twenty naturally-occurring amino acids in proteins can take up positive/negative charge, depending on the pH of the environment. Name any one of the five Aspartic Acid / Aspartate

1 11) Each amino acid in a protein chain contributes three bonds to the polypeptide backbone. Out of these, two are rotatable bonds, and one is not. Answer either (i) which is the non-rotatable bond, by name? or (ii) mention the atoms between which the non-rotatable bond is formed. (i) amide / peptide bond

1 12) If each amino acid is assumed to have three rotatable bonds (two in the polypeptide backbone, and at least one in the sidechain) and if each of these bonds can adopt at least three rotameric states, what is the minimum number of structures that the amino acid can adopt ? $3^3 = 27$

1 13) Hydrogen bonds, hydrophobic interactions and electrostatic interactions can come into play to stabilize a protein of a given sequence in an almost infinite number of ways. (True/False) True

14) Using the twenty naturally-occurring amino acids, in how many different ways (i.e., using how many different amino acid sequences) can you potentially create a protein that is 200 amino acids long? $20^{200} = 10^{400}$

1 15) If an unfolded protein chain that is allowed to begin folding can change its structure roughly once in every microsecond, how many different structures can each chain in a population of folding chains potentially adopt (through random sampling) in a total of one minute of time (assuming no adopted structure is stable)? $60 \times 10^6 = 6\times10^7$

End Semester Exam BIO204-Tissue Regeneration-15 marks; 30 min
(0.25 negative marking for wrong attempt)
Instructor-Rajesh Ramachandran

Name: [illegible]

Roll No: MS24129

15

1. Which process best explains regeneration in Hydra without cell proliferation?
   A. Epimorphosis
   B. Morphallaxis
   C. Fibrosis
   D. Angiogenesis

2. Activation of which pathway is critical for head regeneration in Hydra?
   A. Notch pathway
   B. Hedgehog pathway
   C. Wnt pathway
   D. MAPK pathway

3. Which type of tissue has the highest regenerative capacity?
   A. Permanent tissues
   B. Stable tissues
   C. Labile tissues
   D. Connective tissues

4. Permanent tissues typically respond to injury by:
   A. Complete regeneration
   B. Hyperplasia
   C. Scar formation
   D. Dedifferentiation

5. Which component is essential for proper tissue regeneration due to its structural and signaling role?
   A. DNA
   B. Extracellular matrix (ECM)
   C. Ribosomes
   D. Lysosomes

6. Failure of regeneration in irradiated planarians is due to:
   A. Loss of ECM
   B. Loss of neoblasts
   C. Reduced blood supply
   D. Increased apoptosis

7. Which statement best distinguishes regeneration from repair?
   A. Both always restore original structure
   B. Repair always involves regeneration
   C. Regeneration restores normal structure, repair may cause scarring
   D. Repair is faster than regeneration

8. Stable tissues are best described as:
   A. Continuously dividing
   B. Permanently non-dividing
   C. Quiescent but capable of proliferation
   D. Dead tissues

9. The polarity of Hydra regeneration is primarily governed by:
   A. Neural gradients
   B. Hormonal gradients
   C. Morphogenetic gradients
   D. Mechanical forces

10. Planarian regeneration depends mainly on:
    A. Differentiated epithelial cells
    B. Neurons
    C. Pluripotent stem cells (neoblasts)
    D. Fibroblasts

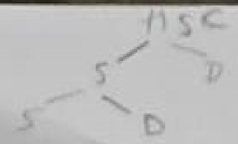
11. Regeneration in mammals is best exemplified by:
    A. Limb regeneration
    B. Brain repair
    C. Liver growth after partial removal
    D. Cardiac muscle repair

12. Which local factor delays healing?
    A. Good nutrition
    B. Adequate oxygen
    C. Poor blood supply
    D. Normal immune function

13. Which cells are primarily responsible for scar formation?
    A. Neutrophils
    B. Fibroblasts
    C. Erythrocytes
    D. Platelets

14. Which of the following is a labile tissue?
    A. Cardiac muscle
    B. Neurons
    C. Intestinal epithelium
    D. Skeletal muscle

15. Keloids are formed due to:
    A. Reduced collagen synthesis
    B. Excessive collagen deposition
    C. Lack of fibroblasts
    D. Increased angiogenesis only



BIO204: End-Semester Examination 2026

INSTRUCTOR: Prof. Lolitika Mandal

Total marls: 20

Please ensure you have returned the question paper along with your answer script.

Answer either set 1 or set 2 from Section A: (10)

A. Rohan is a PhD student whose research is directed in identifying and characterizing novel stem cell population. He has identified a group of cells that he suspects as stem cells. Can you help him in designing two experiments to determine if indeed those cells are really stem cells. (4)

B. Stem cells can undergo both symmetric as well as asymmetric cell division. (4)
Name and explain three mechanisms that controls the balance between these types of divisions?

C. A single hematopoietic stem cell (HSC) undergoes 10 consecutive asymmetric divisions within its niche. What is the total number of stem cells and differentiated (progenitor) cells present after these 10 divisions? (2)

OR

A. If a morphogen concentration drops below the threshold for a specific fate, do cells stop differentiating? (4)
Explain and Justify your answer with an example.

B. Keeping the Waddington model and a stem cell in your consideration, explain at least three factors that can control the epigenetic landscape (3)

C. How do stem cells maintain their identity if the signaling niche moves or is at a distant? (3)
Cite an example.

Answer all in Section B with justification. Just yes or no will not be awarded any marks. (10)

1. Is there a possibility that a niche can function without stem cells? (2)
2. Why can a fertilized egg (zygote) *not* be strictly defined as a stem cell, despite being the ultimate "source" of all cells? (2)

Why is juxtacrine signaling advantageous in development? (2)

Was Dolly a 100% exact genetic copy of the donor sheep? (2)

How can the same signaling pathway (e.g., Wnt) promote stem cell self-renewal in one tissue ...ine) but differentiation in another (skin)? (2)